Furthermore, from the definition of the events $B_1, \ldots, B_n$ it is seen that $B_i \subset A_i$ for $i = 1, \ldots, n$. Therefore, by Theorem 1.5.4, $\Pr(B_i) \le \Pr(A_i)$ for $i = 1, \ldots, n$. It now follows that

$$\Pr\left(\bigcup_{i=1}^{n} A_i\right) \le \sum_{i=1}^{n} \Pr(A_i).$$

(Of course, if the events $A_1, \ldots, A_n$ are disjoint, there is equality in this relation.)

For the second part, apply the first part with A_i replaced by A_i^c for $i = 1, \ldots, n$. We get

$$\Pr\left(\bigcup A_i^c\right) \le \sum_{i=1}^{n} \Pr(A_i^c). \tag{S.1.1}$$

Exercise 5 in Sec. 1.4 says that the left side of (S.1.1) is $\Pr\left(\left[\bigcap A_i\right]^c\right)$. Theorem 1.5.3 says that this last probability is $1 - \Pr\left(\bigcap A_i\right)$. Hence, we can rewrite (S.1.1) as

$$1 - \Pr\left(\bigcap A_i\right) \le \sum_{i=1}^{n} \Pr(A_i^c).$$

Finally take one minus both sides of the above inequality (which reverses the inequality) and produces the desired result.

14. First, note that the probability of type AB blood is $1-(0.5+0.34+0.12) = 0.04$ by using Theorems 1.5.2 and 1.5.3.

    (a) The probability of blood reacting to anti-A is the probability that the blood is either type A or type AB. Since these are disjoint events, the probability is the sum of the two probabilities, namely $0.34 + 0.04 = 0.38$. Similarly, the probability of reacting with anti-B is the probability of being either type B or type AB, $0.12 + 0.04 = 0.16$.

    (b) The probability that both antigens react is the probability of type AB blood, namely 0.04.

1.6 Finite Sample Spaces

Solutions to Exercises

1. The safe way to obtain the answer at this stage of our development is to count that 18 of the 36 outcomes in the sample space yield an odd sum. Another way to solve the problem is to note that regardless of what number appears on the first die, there are three numbers on the second die that will yield an odd sum and three numbers that will yield an even sum. Either way the probability is 1/2.

2. The event whose probability we want is the complement of the event in Exercise 1, so the probability is also 1/2.

3. The only differences greater than or equal to 3 that are available are 3, 4 and 5. These large difference only occur for the six outcomes in the upper right and the six outcomes in the lower left of the array in Example 1.6.5 of the text. So the probability we want is $1 - 12/36 = 2/3$.

4. Let x be the proportion of the school in grade 3 (the same as grades 2–6). Then $2x$ is the proportion in grade 1 and $1 = 2x + 5x = 7x$. So $x = 1/7$, which is the probability that a randomly selected student will be in grade 3.

5. The probability of being in an odd-numbered grade is $2x + x + x = 4x = 4/7$.

6. Assume that all eight possible combinations of faces are equally likely. Only two of those combinations have all three faces the same, so the probability is $1/4$.

7. The possible genotypes of the offspring are aa and Aa, since one parent will definitely contribute an a, while the other can contribute either A or a. Since the parent who is Aa contributes each possible allele with probability $1/2$ each, the probabilities of the two possible offspring are each $1/2$ as well.

8. (a) The sample space contains 12 outcomes: (Head, 1), (Tail, 1), (Head, 2), (Tail, 2), etc.
   (b) Assume that all 12 outcomes are equally likely. Three of the outcomes have Head and an odd number, so the probability is $1/4$.

1.7 Counting Methods

Commentary

If you wish to stress computer evaluation of probabilities, then there are programs for computing factorials and log-factorials. For example, in the statistical software R, there are functions `factorial` and `lfactorial` that compute these. If you cover Stirling's formula (Theorem 1.7.5), you can use these functions to illustrate the closeness of the approximation.

Solutions to Exercises

1. Each pair of starting day and leap year/no leap year designation determines a calendar, and each calendar correspond to exactly one such pair. Since there are seven days and two designations, there are a total of $7 \times 2 = 14$ different calendars.

2. There are 20 ways to choose the student from the first class, and no matter which is chosen, there are 18 ways to choose the student from the second class. No matter which two students are chosen from the first two classes, there are 25 ways to choose the student from the third class. The multiplication rule can be applied to conclude that the total number of ways to choose the three members is $20 \times 18 \times 25 = 9000$.

3. This is a simple matter of permutations of five distinct items, so there are $5! = 120$ ways.

4. There are six different possible shirts, and no matter what shirt is picked, there are four different slacks. So there are 24 different combinations.

5. Let the sample space consist of all four-tuples of dice rolls. There are $6^4 = 1296$ possible outcomes. The outcomes with all four rolls different consist of all of the permutations of six items taken four at a time. There are $P_{6,4} = 360$ of these outcomes. So the probability we want is $360/1296 = 5/18$.

6. With six rolls, there are $6^6 = 46656$ possible outcomes. The outcomes with all different rolls are the permutations of six distinct items. There are $6! = 720$ outcomes in the event of interest, so the probability is $720/46656 = 0.01543$.

7. There are 20^{12} possible outcomes in the sample space. If the 12 balls are to be thrown into different boxes, the first ball can be thrown into any one of the 20 boxes, the second ball can then be thrown into any one of the other 19 boxes, etc. Thus, there are $20 \cdot 19 \cdot 18 \cdots 9$ possible outcomes in the event. So the probability is $20!/[8!20^{12}]$.